Plant Disease Classification

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1. Problem Statement and Scope

The failure to detect plant diseases in their early stages can result in a rapid spread across farmland and inflict major economic losses. Farmers use their experience and expertise to diagnose diseases based on visual symptoms, including discoloration of leaves, the presence/absence of spots on leaves, or changes in the texture of leaves. This method, however, is inherently flawed, as the accuracy of the diagnosis is often subjective, varying from person to person, and is often prone to error.

This project aims to develop a method or a model based on machine learning and artificial intelligence that can autonomously identify plant diseases based on images of the leaves of the plants. This model will autonomously classify images into a number of predetermined, identified, and cataloged categories of diseases using a convolutional neural network (CNN). The work in this project will include training and testing the classification model and testing the model against the explainable AI model, as well as the integration of the AI model and the RL model.

The focus of the work will be on the classification of plant diseases, and in particular, the use of images for the diagnosis of plant diseases. The work, however, will not be intended to replace the professional diagnosis of plant diseases in the field of agriculture. The work will serve as a decision-support tool to aid farmers and people who work in the field of agriculture in the identification of potential plant diseases.

1. Related Work

Plant diseases significantly affect agricultural productivity and crop yield worldwide. Traditional detection methods rely on manual inspection, which can be time-consuming and prone to human error. These limitations have encouraged the use of machine learning and deep learning techniques to improve the accuracy and efficiency of plant disease detection [1].

Recent studies have shown that deep learning techniques, particularly convolutional neural networks (CNNs), are effective for detecting plant diseases using leaf images. These models can automatically learn visual features such as discoloration, spots, and texture differences that indicate the presence of plant diseases. CNNs can automatically extract important features from raw image data, enabling more accurate and efficient classification of plant diseases based on visible symptoms [2]. Because of this capability, CNN-based systems are increasingly used in agricultural technology to support disease identification.

Public image datasets are commonly used in developing plant disease detection models, particularly the PlantVillage dataset, which contains a large collection of labeled plant leaf images. In one study, the PlantVillage dataset was used to train several convolutional neural network models, where the VGG16 model achieved the best performance with an accuracy of 90.4% [3]. The results demonstrate that deep CNN architectures trained on large image datasets can effectively classify plant diseases and support automated disease identification systems.

Similarly, another study applied the EfficientNet deep learning architecture for classifying plant leaf diseases. The results showed that the EfficientNet B5 and B4 models achieved very high performance, reaching 99.91% and 99.97% accuracy on the original and augmented datasets, respectively [4]. These findings demonstrate that deep learning models are highly effective in recognizing patterns associated with plant diseases.

Despite the high accuracy reported in previous studies, several limitations remain in plant disease detection systems. Many models are trained using controlled datasets such as the widely used PlantVillage dataset, where images are captured under consistent lighting conditions and simple backgrounds. However, real-world plant images often contain complex backgrounds, varying lighting conditions, and different leaf orientations, which can make segmentation of affected areas more difficult and may reduce the accuracy of disease classification in practical applications [5].

These studies demonstrate that CNN-based image classification systems are effective tools for detecting plant diseases. Building on these studies, the proposed research aims to develop a CNN-based plant disease classification model capable of automatically identifying diseases from leaf images in practical agricultural settings.

2. Dataset Plan

The study will utilize the **PlantVillage dataset**, a publicly available dataset that contains thousands of labeled images of both healthy and diseased plant leaves. This dataset is publicly available through research repositories such as Kaggle and the official PlantVillage project website. This includes multiple plant species and various categories of plant diseases, making it suitable for supervised image classification tasks using convolutional neural networks (CNNs). Each image in the dataset is labeled according to the plant species and the corresponding disease condition or healthy state, allowing the model to learn and classify patterns associated with specific plant diseases.

The PlantVillage dataset has been widely used in research related to plant disease detection and agricultural artificial intelligence systems. Because of its large number of labeled images and clearly defined disease categories, it provides a reliable dataset for training and evaluating machine learning models designed for plant disease classification. The dataset is publicly available for research and academic use and does not contain proprietary or restricted data.

Since the dataset contains only images of plant leaves and does not include any human subjects, personal information, or identifiable data, no individual consent is required for its use in this research. Despite its advantages, the dataset also has certain limitations. Many of the images were captured under controlled conditions with uniform backgrounds and stable lighting. As a result, the dataset may not fully represent real-world agricultural environments where images can vary significantly in terms of lighting conditions, background noise, camera angles, and environmental factors.

To address these limitations, the study will apply several **data augmentation techniques** during the training process. These techniques include image rotation, horizontal and vertical flipping, scaling, and brightness adjustments. Data augmentation increases the diversity of the training data and helps improve the model's ability to generalize when applied to real-world scenarios.

Furthermore, the dataset will undergo preprocessing steps such as **image resizing, normalization of pixel values, and splitting into training, validation, and testing sets**. These steps are essential to ensure consistent input for the CNN model and to evaluate its performance effectively. Although the dataset is large and well-structured, the study acknowledges the potential presence of dataset bias and recognizes that generalization to real-world agricultural environments may still have limitations.

Overall, the PlantVillage dataset provides a strong foundation for developing a machine learning model capable of identifying plant diseases from leaf images while maintaining ethical standards since the dataset contains no personal or identifiable information.

3. Planned CNN / NLP / RL Components and MVP

The primary component for this project consists of a **convolutional neural network (CNN)** designed to classify plant diseases using images of plant leaves. The MobileNetV3 architecture, a lightweight and efficient CNN model optimized for image classification, will be used [6]. Transfer learning will be applied using pretrained weights from the ImageNet database. The model will then be fine-tuned using the PlantVillage dataset.

To increase the model's interpretability, Grad-CAM, which is a technique that produces visualizations to explain CNN model decisions, will also be integrated [7]. Specifically, the Grad-Cam will highlight specific areas of a leaf image that the model used to make a prediction. Providing these visual explanations helps users identify if the model actually focuses on relevant features on the inputted images. This improves model transparency and allows technical users to have a better understanding of the model's process of identifying diseases.

The project will include a lightweight component of **natural language processing (NLP)** to provide simple and clear text explanations for model predictions. The model may generate short descriptions such as: "Predicted disease: Leaf Blight with 92% confidence." The purpose of this NLP module is to provide a more user-friendly explanation to users of the model's predictions. This makes results easier to interpret for non-technical users such as farmers or vendors that may not be familiar with machine learning techniques.

A minimal **reinforcement learning (RL)** component will also be incorporated during training to experiment and find improvements in data augmentation and image preprocessing [8]. The RL agent will explore various configurations and adjustments in image factors such as brightness levels, scaling, and angles. Following each improvement in validation accuracy, the agent will receive a corresponding reward. This encourages the agent to prioritize strategies that will lead to high validation scores in future training iterations.

The project's **minimum viable product (MVP)** will consist of a trained MobileNetV3-based plant disease classification model that accepts leaf images as input and outputs the predicted disease label, confidence score, Grad-CAM visualization, and a simple textual explanation. The model's performance will be evaluated using standard machine learning evaluation metrics such as accuracy, macro-F1, and per-class confusion matrix.

4. Ethics Risk Register

A major consideration for automated plant disease detection systems is misdiagnosis and its consequences. Misdiagnosis is predicting a disease that a farmer's crop does not have and can lead to economic or environmental problems. To lessen the risks of misdiagnosis, the system will include prediction confidence scores and disclaimers stating that the system is designed to assist users in the treatment decision-making process.

A further risk of automated plant disease detection systems is dataset bias. Training datasets might not cover different plant types, the potential range of environmental conditions, or the different variations of disease and pathogens; therefore, the model's effectiveness may be limited in real environments. To lessen the risks, model training will include a range of differing conditions, and the project will evaluate the model's effectiveness in the different disease categories of the training dataset.

Another ethical concern is the potential over-reliance on automated AI systems. Farmers or users may rely too heavily on AI predictions without consulting agricultural experts, which could lead to incorrect treatment decisions if the model produces inaccurate predictions. To mitigate this risk, the system will clearly indicate that the predictions are advisory and should be used together with professional agricultural guidance rather than as a replacement for expert diagnosis.

5. Team Roles & Milestones

The project team will divide responsibilities to ensure efficient development and completion of the project. The project timeline will follow the milestones defined in the course schedule.

Roles

1. Magat Jr., Gabriel B. - Evaluation & MLOps Lead
2. Limpin, Paulo Miguel R. - Evaluation & MLOps Lead
3. Alfaro, Javie T. - Modeling Lead
4. Lara, Matt Christian R. - Project Lead / Integration
5. Vender Joalnes John D. - Data & Ethics Lead

Milestone

Week 1: Proposal & GitHub setup (v0.1)

6. References

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